

# Project Report

Heart Disease Analysis

|              |                        |
|--------------|------------------------|
| Team ID      | SWTID-2026-9948        |
| Project Name | Heart Disease Analysis |
| Date         | 12-07-2026             |

# 1. Introduction

## 1.1 Project Overview

Heart Disease Analysis is an interactive Tableau dashboard project built to explore and visualize the relationship between heart disease, stroke, and a range of demographic, lifestyle, and health indicators. Using a BRFSS-style health survey dataset containing fields such as Sex, Race, Age Category, Diabetic Status, Smoking, Alcohol Drinking, General Health, and BMI, the dashboard presents multiple linked visuals (bar charts, pie chart, bubble chart, scatter plot, and a heat grid) across two dashboard pages, connected through a guided "story" flow with Previous/Next navigation.

## 1.2 Purpose

The purpose of this project is to give healthcare analysts, public health planners, and decision makers a quick, visual way to identify which population groups are most at risk of heart disease and stroke. By combining demographic filters (Age, Gender, Race) with lifestyle factors (Smoking, Alcohol Drinking) and health indicators (BMI, General Health, Diabetic Status), the dashboard aims to surface actionable patterns that would be difficult to spot in raw tabular data, and to support targeted awareness and prevention efforts.

# 2. Ideation Phase

## 2.1 Problem Statement

Heart disease and stroke remain among the leading causes of death worldwide, yet the risk factors behind them are spread across many disconnected data points - demographics, lifestyle habits, and underlying health conditions. Health workers and analysts often lack a single, easy-to-use visual tool to correlate these factors and quickly identify which groups need the most attention, making early risk identification slower and less data-driven than it could be.

## 2.2 Empathy Map Canvas

**User group:** Public health analysts / healthcare decision makers

|                       |                                                                                                                                    |
|-----------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>Says / Does</b>    | Requests clear, filterable views of risk factors; frequently cross-checks age, race and lifestyle data before drawing conclusions. |
| <b>Thinks / Feels</b> | Concerned about missing at-risk groups; feels overwhelmed by scattered spreadsheets and static reports.                            |
| <b>Sees</b>           | Large raw survey datasets, existing static PDF health reports, dashboards from other departments.                                  |
| <b>Hears</b>          | Requests from leadership for quick answers on "who is most at risk" and "what factors matter most".                                |
| <b>Pain</b>           | Time-consuming manual analysis; difficulty correlating multiple risk factors simultaneously.                                       |
| <b>Gain</b>           | An interactive, filterable dashboard that answers risk-group questions in seconds.                                                 |

## 2.3 Brainstorming

Ideas considered before finalizing the approach:

- A static PDF/Excel report summarizing risk factors (rejected - not interactive, hard to update).
- A machine learning model to predict individual heart disease risk (parked for future scope - requires more granular/labelled data and longer development time).
- A mobile app for individual self-risk checks (rejected for this phase - out of scope for an analytics-focused deliverable).
- An interactive BI dashboard (Tableau) with cross-filterable visuals and a guided story - selected as the best fit for quick, explorable population-level insights.

## 3. Requirement Analysis

### 3.1 Customer Journey Map

- **Awareness:** Analyst receives the raw heart disease/stroke survey dataset.
- **Access:** Analyst opens the published Tableau dashboard link.
- **Exploration:** Analyst applies filters (Age Category, Gender, Race, Smoking, Alcohol Drinking, Diabetic Status) to narrow down a population segment.
- **Insight:** Analyst reviews the relevant charts (e.g. Age & BMI vs Diabetic, Smoking & Alcohol Risk Factors) to identify high-risk patterns.
- **Action:** Analyst shares findings/screenshots with stakeholders or exports the story for a report/presentation.

### 3.2 Solution Requirement

#### Functional Requirements:

- Display counts of Heart Disease and Stroke across demographic and lifestyle segments.
- Provide interactive filters for Age Category, Gender, Race, Smoking, Alcohol Drinking, Diabetic Status, and General Health.
- Show Average BMI trends across Age Category and Diabetic Status.
- Provide a guided, multi-page story with navigation controls.

#### Non-Functional Requirements:

- Dashboard should load and respond to filter changes within a few seconds.
- Visuals must remain accurate and legible when combined with multiple filters.
- The dashboard should be accessible without requiring specialized software (Tableau Public / browser-based viewing).

### 3.3 Data Flow Diagram

Raw Survey Dataset (CSV) → Data Cleaning & Preprocessing (handling nulls/duplicates, standardizing Yes/No fields, binning Age & BMI) → Tableau Data Source / Data Model (calculated fields for Count of Heart Disease, Count of Stroke, Average BMI, % by Race) → Dashboard Visuals (bar, pie, bubble, scatter, heat grid) → Guided Story (Page 1 & Page 2) → End User (filters,

explores, and draws insights).

### 3.4 Technology Stack

| Category                     | Technology                                                    |
|------------------------------|---------------------------------------------------------------|
| Data Visualization / BI Tool | Tableau (Public/Desktop)                                      |
| Data Storage / Input         | CSV / Excel (BRFSS-style health survey dataset)               |
| Data Preparation             | Tableau data source pane, calculated fields, grouping/binning |
| Hosting / Sharing            | Tableau Public workbook link                                  |
| Documentation                | Google Docs / Microsoft Word (project templates), PDF reports |

## 4. Project Design

### 4.1 Problem Solution Fit

Analysts need a fast way to correlate multiple risk factors for heart disease and stroke, but existing static reports don't allow real-time cross-filtering. An interactive Tableau dashboard directly fits this need: it lets users combine any set of demographic, lifestyle, and health filters and immediately see updated visuals, closing the gap between raw data and actionable insight.

### 4.2 Proposed Solution

A two-page Tableau dashboard connected as a guided story: Page 1 focuses on lifestyle and general health risk factors (Smoking & Alcohol Risk Factors, General Health vs Heart Disease, Age & BMI vs Diabetic, Heart Disease/Diabetic Status/Age Category breakdown); Page 2 focuses on demographic risk factors (Gender vs Heart Disease, Diabetic vs Heart/Stroke, Race-wise Heart Disease, and a repeated Smoking & Alcohol view for cross-reference). Shared legends and filters (Stroke, Age Category, Gen Health, Diabetic Status, Race) keep both pages synchronized.

### 4.3 Solution Architecture

The architecture follows a simple BI pipeline: (1) Data Layer - the cleaned CSV dataset; (2) Modeling Layer - Tableau data source with calculated fields (Count of Heart Disease, Count of Stroke, Average BMI, Race percentage); (3) Presentation Layer - two interconnected dashboard pages built from multiple chart types, sharing a common set of filters/legends; (4) Delivery Layer - the dashboard is published as a Tableau story with Previous/Next navigation for end users to consume via a browser.

## 5. Project Planning & Scheduling

### 5.1 Project Planning

| Phase                            | Activities                                                | Duration |
|----------------------------------|-----------------------------------------------------------|----------|
| Ideation Phase                   | Problem statement, empathy map, brainstorming             | 2 days   |
| Requirement Analysis             | Customer journey map, requirements, data flow, tech stack | 2 days   |
| Project Design                   | Problem-solution fit, proposed solution, architecture     | 2 days   |
| Project Development              | Data cleaning, dashboard/story development in Tableau     | 7 days   |
| Functional & Performance Testing | Functional testing, load-time and UAT testing             | 2 days   |
| Documentation & Submission       | Final report, screenshots, sign-off                       | 1 day    |

## 6. Functional and Performance Testing

### 6.1 Performance Testing

| S.No. | Parameter               | Screenshot / Values                                                                                                                                                     |
|-------|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.    | Data Rendered           | Dataset fields (Sex, Race, Age Category, Diabetic Status, Smoking, Alcohol Drinking, General Health, BMI, Heart Disease, Stroke) rendered correctly across all visuals. |
| 2.    | Data Preprocessing      | Nulls/duplicates removed; Yes/No fields standardized; Age binned into Age Category; Average BMI computed (23.91 - 42.78); General Health grouped into 5 levels.         |
| 3.    | Utilization of Filters  | Filters on Age Category, Gender, Race, Diabetic Status, Smoking, Alcohol Drinking, General Health, Heart Disease and Stroke.                                            |
| 4.    | Calculation fields Used | Count of Heart Disease, Count of Stroke, Count of Id, Average BMI, % of Heart Disease by Race.                                                                          |
| 5.    | Dashboard design        | No. of Visualizations/Graphs - 11 across 2 dashboard pages.                                                                                                             |
| 6.    | Story Design            | No. of Visualizations/Graphs - 2 dashboard pages combined into a guided story.                                                                                          |

(See the separate Performance Testing and UAT reports for full functional test case details.)

## 7. Results

### 7.1 Output Screenshots

The final dashboard consists of two connected pages published as a Tableau story:

- **Page 1 - Lifestyle & Health Risk Factors:** Smoking & Alcohol Risk Factors bar chart, General Health vs Heart Disease bubble chart (Poor 92, Fair 140, Good 172), Age & BMI vs Diabetic heat grid (Avg. BMI range 23.91 - 42.78), and a Heart Disease/Diabetic Status/Age Category clustered bar chart.
- **Page 2 - Demographic Risk Factors:** Gender vs Heart Disease bar chart (Female/Male counts), Diabetic vs Heart/Stroke scatter plot, Race-wise Heart Disease pie chart (White 75.57%, Black 17.81%, Hispanic 3.73%, remainder Asian/Other/American Indian), and a repeated Smoking & Alcohol Risk Factors view.
- Both pages share a common legend panel (Stroke, Age Category, Gen Health, Diabetic Status, Race, Avg. BMI) and are linked via Previous/Next navigation.

[Insert dashboard screenshots here for the final submission.]

## 8. Advantages & Disadvantages

### Advantages:

- Provides quick, visual identification of high-risk population groups.
- Interactive filters let users explore any combination of demographic/lifestyle/health factors.
- Accessible to non-technical stakeholders without requiring coding or statistical knowledge.
- Free to build and share via Tableau Public.

### Disadvantages:

- Descriptive only - does not predict individual risk (no ML model).
- Relies on the accuracy and completeness of the underlying survey dataset.
- Requires an internet connection and browser to view a Tableau Public dashboard.
- Data is not real-time; the dashboard reflects a static, point-in-time dataset.

## 9. Conclusion

The Heart Disease Analysis dashboard successfully brings together demographic, lifestyle, and health data into a single, interactive Tableau story. It allows analysts to quickly filter and cross-reference risk factors such as smoking, alcohol consumption, BMI, diabetic status, race, and gender against heart disease and stroke outcomes, turning a large raw dataset into clear, actionable visual insights.

## 10. Future Scope

- Add a predictive Machine Learning model to estimate individual heart disease/stroke risk scores.
- Integrate real-time or periodically refreshed health survey data instead of a static dataset.
- Expand demographic granularity (e.g. income, geography/state-level breakdowns).

- Build a mobile-friendly or embedded version for wider public/clinical use.

## 11. Appendix

**Source Code (if any):** Not applicable - built using Tableau's visual interface (no custom source code); Tableau workbook (.twbx) available on request.

**Dataset Link:** [Insert link to the BRFSS-style Heart Disease survey dataset used]

**GitHub & Project Demo Link:** [Insert GitHub repository link and Tableau Public dashboard/demo link]